

Verification of the solution $(1, 0, -1)$

$$1 - 2(0) + (-1) = 1 - 0 + (-1) = 0 = \text{RHS}$$

$$2(0) - 8(-1) = 0 + 8 = 8 = \text{RHS}$$

$$5(1) - 5(-1) = 5 + 5 = 10 = \text{RHS}$$

The results agree with the right side of the original system, so $(1, 0, -1)$ is a solution of the system. $\square$

ELEMENTARY ROW OPERATIONS

1. (Replacement) Replace one row by the sum of itself and a multiple of another row.
2. (Interchange) Interchange two rows.
3. (Scaling) Multiply all entries in a row by a nonzero constant.

Two matrices are called row equivalent if there is a sequence of elementary row operations that transforms one matrix into the other.

It is important to note that row operations are reversible. If two rows are interchanged, they can be returned to their original positions by another interchange. If a row is scaled by a nonzero constant c , then multiplying the new row by $1/c$ produces the original row. Finally, consider a replacement operation involving two rows — say, rows 1 and 2 — and suppose that c times row 1 is added to row 2 to produce a new row 2. To "reverse" this operation, add $-c$ times row 1 to (new) row 2 and obtain the original row 2.

Suppose a system is changed to a new one via row operations. By considering each type of row operation, you can see that any solution of the original system remains a solution of the new system. Conversely, since the original system can be produced via row operations on the new system, each solution of the new system is also a solution of the original system.

This discussion justifies the following statement: